

Anand Patel

anandpatel1221@icloud.com | linkedin.com/in/anand-patel1221 | github.com/anand1221178 | [Portfolio](#)

EDUCATION

University of Illinois Urbana-Champaign

Urbana-Champaign, IL

Master of Computer Science

Expected May 2028

- **Advanced Coursework:** Advanced Distributed Systems, Advanced Operating Systems, Parallel Computer Architecture, Advanced Compiler Construction

University of the Witwatersrand

Johannesburg, South Africa

BSc Honours in CS, GPA 4.0/4.0 – Distinction | PG Merit Award, Most Innovative Research

Graduated Dec. 2025

University of the Witwatersrand

Johannesburg, South Africa

BSc in Computer Science, GPA 3.91/4.0 | Certificate of Merit in Algorithms, Dean's List

Graduated Dec. 2024

PUBLICATIONS

PEARC'26 | Practice and Experience in Advanced Research Computing

Accepted 2026

- *Extending the Life of HPC Systems in Resource Constrained Environments: Mapping Productivity-Energy Trade-offs in Memory-Bound Workloads via DVFS and Core Scaling on Repurposed Hardware.* **Best Student Short Paper**, Track: Systems/Infrastructure/Middleware. Led experimental design, DVFS methodology, and the performance analysis framework

EXPERIENCE

Performance Engineer

Aug. 2025 – Jun. 2026

Private BlockDAG Startup

Johannesburg, South Africa

- Owned end-to-end performance engineering for production Redis infrastructure; diagnosed live production issues and tuned distributed caching to improve latency, throughput, and memory utilization under real load

Team Lead & Mentor, Student Cluster Competition

Jan. 2023 – Jan. 2025

Centre for High Performance Computing (CHPC)

Johannesburg, South Africa

- Architected and operated a distributed Linux cluster from bare metal, running large-scale workloads (HPL, HPCG) across the fleet under a strict 3kW power constraint, placing **2nd overall in Africa**
- Engineered automated log analysis pipelines and Grafana/Prometheus telemetry for real-time visibility into cluster health and failure modes; awarded **Best Male Performer in Africa (2023)**

Junior Software Developer & AWS Cloud Captain

Sep. 2023 – Dec. 2024

University of the Witwatersrand

Johannesburg, South Africa

- Designed scalable backend services in **Java** and Spring Boot and optimized database queries for concurrent request handling; worked in large Agile teams across CI/CD pipelines and code review, and led cloud workshops as AWS Cloud Captain

High Performance Computing Intern

Dec. 2023

Council for Scientific and Industrial Research (CSIR)

Pretoria, South Africa

- Optimized distributed storage and networking for Square Kilometre Array (SKA) infrastructure, addressing high-volume data ingest and large-scale data movement from the telescope array

PROJECTS

eBPF/XDP ITCH Market Data Parser | C, eBPF, XDP, Linux Kernel

2026

- Built a low-latency NASDAQ ITCH 5.0 market data ingest pipeline parsing the full Ethernet/IP/UDP/MoldUDP64/ITCH stack at the NIC driver level, bypassing the kernel network stack
- Achieved **930 ns** average per-packet latency (bpftool-measured) across 500K+ packets, with BPF hash map symbol filtering and zero-copy ring buffer delivery into a lock-free C++ order book maintaining live best bid/ask

TopoSteal – NUMA-Aware Work-Stealing Runtime | C, Linux, hwloc, perf

2026

- Built a topology-aware parallel runtime that discovers cache topology via hwloc and reweights work-stealing decisions using live hardware PMU counters (perf_event_open) to bias steals toward cache-local victims
- Achieved **1.17× speedup** and 85% same-socket locality over random stealing (5 configs, 50 trials); lock-free Chase-Lev dequeues with C11 atomics stress-tested to 81M tasks, outperforming OpenMP static scheduling by 5% on a 77M-nonzero SuiteSparse SpMV workload

Sentinel RTOS | C, ARM Assembly, STM32F401RE

2026

- Designed a fully preemptive RTOS from scratch on bare metal – no HAL, no FreeRTOS – with a custom priority scheduler with priority inheritance, PendSV-driven context switching in ARM assembly, and synchronisation primitives
- Wrote bare-metal drivers for UART, SPI, I2C, PWM, GPIO, and ADC via register manipulation; built MicroBench to benchmark context-switch latency and interrupt response against FreeRTOS

Autonomous Robotic Failure Recovery (Honours Thesis) | Python, PyTorch, RL

2025

- Developed an RL policy for quadruped locomotion under simulated actuator failures and sensor drift, formally characterizing negative gradient interference between SR2L and Domain Randomization across 38,000 parallelized episodes. Also built a CUDA ray tracer hitting 187M rays/sec via warp-level parallelism and coalesced memory access

TECHNICAL SKILLS

Languages: Java, C/C++ (C++20, atomics, lock-free), Python, SQL, Bash, Kotlin, CUDA, ARM Assembly

Distributed & Infra: Distributed Storage, Linux Clustering, MPI, Redis, Spring Boot, REST APIs, CI/CD, Grafana/Prometheus

Systems & OS: Linux Kernel Modules, Device Drivers, Custom Schedulers, pthreads, eBPF/XDP, NUMA-aware Scheduling

Tools: Git, CMake, GDB, Valgrind, perf, Nsight, gem5, STM32, PyTorch, OpenMP